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Abstract	The global real estate market faces inefficiencies, fraud risks, and high entry barriers. This research presents DeFiHomes, a decentralized property marketplace and bidding platform leveraging blockchain technology to streamline transactions, eliminate intermediaries, and ensure secure, transparent ownership transfers. By tokenizing properties as NFTs on the Ethereum blockchain and using smart contracts for auctions and title transfers, the platform reduces reliance on central authorities and addresses issues like documentation fraud and ownership disputes. The solution integrates web frameworks (React, Next.js) with decentralized storage (IPFS) and secure wallet authentication (MetaMask). It features real time bidding, auction countdowns, and bid tracking. It manages off-chain data like user profiles and bid analytics with row-level security. The paper evaluates the architecture's effectiveness in enhancing transparency, lowering investment thresholds via fractional ownership potential, and providing tamper-resistant records. This platform aims to set a benchmark for future real estate innovations, prioritizing trust, efficiency, and inclusivity.

Keywords (separated by '-')	Real Estate - Blockchain - Property Tokenization - NFT - Smart Contracts - Decentralized Application - Ethereum - IPFS - MetaMask

DeFiHomes: A Secure and Transparent Decentralized Real Estate Platform with Smart Contract-Driven Auctions

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Abstract. The global real estate market faces inefficiencies, fraud risks, and high entry barriers. This research presents DeFiHomes, a decentralized property marketplace and bidding platform leveraging blockchain technology to streamline transactions, eliminate intermediaries, and ensure secure, transparent ownership transfers. By tokenizing properties as NFTs on the Ethereum blockchain and using smart contracts for auctions and title transfers, the platform reduces reliance on central authorities and addresses issues like documentation fraud and ownership disputes. The solution integrates web frameworks (React, Next.js) with decentralized storage (IPFS) and secure wallet authentication (MetaMask). It features real time bidding, auction countdowns, and bid tracking. It manages off-chain data like user profiles and bid analytics with row-level security. The paper evaluates the architecture's effectiveness in enhancing transparency, lowering investment thresholds via fractional ownership potential, and providing tamper-resistant records. This platform aims to set a benchmark for future real estate innovations, prioritizing trust, efficiency, and inclusivity.

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1 Introduction

The global real estate industry, valued at over $326 trillion, suffers from opacity, fraud, high costs, and bureaucratic inefficiencies due to reliance on fragmented legacy systems and multiple intermediaries [1]. Persistent challenges include title disputes, lack of standardization, limited asset liquidity, and high barriers to entry, impeding equitable participation and innovation. Decentralized technologies, particularly blockchain, smart contracts, tokenization protocols, and distributed storage, offer a paradigm shift to address these long-standing issues [2].

This paper introduces DeFiHomes, a fully decentralized real estate marketplace designed to overcome these challenges. The platform integrates NFT based property tokenization (ERC-721 standard), smart contract-governed auction systems, decentralized document storage via the Interplanetary File System (IPFS), and secure wallet-based authentication using MetaMask. By representing properties as unique NFTs on

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the Ethereum blockchain (or compatible L2s), with metadata cryptographically linked to verifiable documents on IPFS, DeFiHomes establishes an immutable and transparent record of ownership and provenance, replacing vulnerable traditional paper-based systems [3].

DeFiHomes directly addresses critical friction points such as documentation fraud, escrow delays, and investment inaccessibility through a modular smart contract architecture. This architecture comprises distinct contracts for Property Registry, Auction Engine, Escrow Management, and Ownership Transfer, automating complex workflows and reducing reliance on centralized authorities like title agencies, brokers, and escrow services. The platform architecture inherently supports both full property sales and lays the groundwork for fractional ownership models (e.g., using ERC-1155 extensions), potentially enhancing asset liquidity and broadening investor access.

The user-facing application is built using modern web technologies (React, Next.js, TailwindCSS) for a responsive and performant experience. Supabase provides backend services for managing essential off chain data, such as user profiles and detailed analytics, while employing Row-Level Security (RLS) for granular access control. Secure user authentication is handled via MetaMask, leveraging cryptographic signatures instead of traditional credentials, aligning with Web3 security principles.

Recognizing the importance of visualization in real estate, the platform integrates the Google Maps API for geospatial context and supports embedded YouTube virtual tours. Sellers can manage listings, upload multimedia content, and define auction parameters (reserve price, duration, increments) through a unified dashboard. All significant on-chain events (bid placements, auction finalization, NFT transfers) are monitored via blockchain event listeners and reflected in the user interface in real-time.

A key contribution is the development of a real time analytics dashboard, powered by Chart.js and Supabase subscriptions. This provides users with interactive visualizations of ongoing bids, market valuations, historical transaction data, and portfolio performance, enabling data-driven investment decisions without intermediary bias.

Architectural considerations include scalability, ad dressed through the use of upgradeable proxy contracts (e.g., UUPS pattern), and security, ensured via rigorous testing, static analysis (e.g., Slither, Mythril), and adherence to best practices like reentrancy guards. Critically, the platform operates non-custodially; user funds and assets are managed directly by smart contracts and user wallets, minimizing counterparty risk. DeFiHomes demonstrates a practical framework for leveraging decentralized technologies to foster transparency, security, efficiency, and inclusivity in the real estate market, setting a precedent for future innovations in property tokenization, micro-investment, and programmable ownership.

2 Related Work

Traditional real estate transactions are notoriously inefficient, relying on centralized verification, fragmented records, and manual coordination. Blockchain technology has emerged as a foundational layer for reform. Early pilots, such as Sweden's Lantm ateriet-land registry exploration and the Republic of Georgia's title management system, demonstrated potential reductions in fraud and transaction times [4]. The concept of replacing paper deeds with cryptographic tokens gained traction, leveraging public ledgers for provenance [12].

Non-Fungible Tokens (NFTs), particularly ERC-721 and ERC-1155 standards, have become key enablers for representing unique real-world assets like property on-chain. Platforms like Propy and RealT pioneered tokenizing physical properties, linking NFTs to legal documents and ownership history [3]. This approach enhances immutability and global verifiability, mitigating title fraud [5]. Our work extends this by integrating NFTs with verifiable decentralized metadata storage via IPFS.

Smart contracts automate complex agreements, re moving intermediaries like escrow agents and brokers [6, 7]. Research has explored Solidity-based systems for automating auctions, title issuance, rental agreements, and escrow [8, 13]. DeFiHomes adopts a modular smart contract architecture (Registry, Auction, Es crow, Transfer) optimized for gas, security (reentrancy guards), and upgradeability (proxy patterns).

Decentralized identity (DID) frameworks and wallet-based authentication are crucial for secure, user-controlled interactions in DApps. Tools like Meta Mask allow users to interact with smart contracts via cryptographic signatures, replacing insecure user name/password systems [9]. DeFiHomes leverages MetaMask for secure authentication and transaction signing.

Geospatial accuracy and visualization are vital. Integrating mapping APIs (Google Maps, OpenStreetMap) and virtual tour capabilities enhances user understanding of property context and attributes [11, 14]. DeFiHomes incorporates Google Maps and YouTube embeds for this purpose.

Fractional ownership, enabled by token standards like ERC-1155 or specialized vault protocols, ad dresses the high capital barrier in real estate, potentially unlocking liquidity and enabling micro investments. While DeFiHomes focuses on full property auctions initially, its architecture is designed to accommodate future fractionalization.

3 Problem Definition

The real estate industry continues to struggle with inefficiencies such as high intermediary costs, de lays, and risks of fraud due to fragmented and non-transparent processes. Traditional transactions often require multiple third parties, leading to increased complexity and limited trust, especially in cross-border deals. Existing digital platforms fall short in sup porting secure, real-time bidding, automated escrow, and verifiable ownership transfers, while also lacking privacy due to centralized data models. This research proposes a decentralized platform that addresses these issues through blockchain technology—using NFT based property representation, smart contracts for se cure and automated execution, decentralized storage 3 for media assets, and MetaMask for secure user authentication—offering a streamlined, scalable, and tamper-resistant approach to property transactions.

4 Methodology

DeFiHomes is designed as a decentralized application (DApp) that combines on-chain smart contracts for core logic and off-chain components for user interface, data management, and enhanced features as shown in Fig. 1.

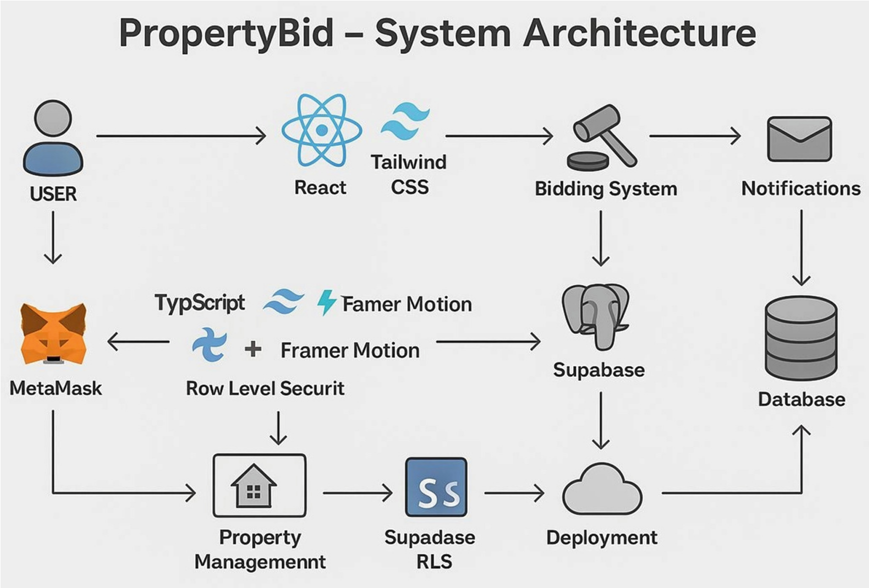

Fig. 1. System Architecture and Technical Stack

4.1 NFT-Based Property Tokenization (ERC 721)

Each unique real estate property listed on the plat form is minted as an ERC-721 NFT on an EVM compatible blockchain (e.g., Ethereum mainnet or a Layer 2 like Polygon). The ERC-721 standard ensures uniqueness and non-fungibility, suitable for representing distinct assets. The NFT's metadata includes:

- A unique property identifier.
- Essential descriptive attributes (address, size, type, etc.).
- Geospatial coordinates.
- A link (typically an IPFS CID) to a JSON file containing more detailed off-chain metadata and links to associated documents.

This token serves as the digital representation of ownership, transferable on the blockchain.

4.2 IPFS for Decentralized Document Storage

Given the high cost and inefficiency of storing large files on-chain, all supplementary documents (legal deeds, inspection reports, floor plans, high-resolution images, virtual tour videos) are stored off-chain using IPFS [10]. Upon listing, documents are uploaded to IPFS, generating a unique, immutable content identifier (CID) based on the file's content. CID is referenced by the NFT's URI stored on-chain. This CID is then stored within the off-chain metadata JSON file, which itself is referenced by the NFT. This ensures:

- **Decentralized Persistence:** Files are stored across the IPFS network, not on a single server.
- **Tamper Evidence:** Any modification to a document changes its CID, making tapering easily detectable.
- **Cost Efficiency:** Avoids expensive blockchain storage fees.

Availability is ensured through pinning services that guarantee nodes continue to host the data.

4.3 Smart Contract Architecture for Auction and Escrow

We've put together a versatile collection of audited Solidity smart contracts aimed at powering a decentralized property auction system. Each contract adheres to Open Zeppelin's upgradeable proxy standard, which means it's built to scale and can be upgraded down the line. This ensures long-term maintainability and compatibility with evolving blockchain standards.

PropertyRegistry.sol: Records on-chain property identities and metadata hashes.
AuctionHouse.sol: Handles live auctions, bid sub missions, and real-time bid validation.
EscrowVault.sol: Temporarily holds bidder funds, releasing them upon auction completion or refunding on bid loss.
OwnershipTransfer.sol: Automates NFT reassignments after escrow fulfillment.

Each smart contract follows OpenZeppelin's upgradeable proxy standard to support version control and ensure future scalability as shown in Fig. 2.

4.4 Web3 Frontend and User Interaction

The frontend is built with React and Next.js, inter acting with the blockchain via libraries like Ethers.js or Web3.js and a user's connected Web3 wallet (Meta Mask). Key interactions include:

- **Wallet Connection:** Users connect their wallet to identify themselves and interact with the DApp.
- **Property Listing:** Sellers initiate a transaction to mint an NFT via the Property Registry contract, providing metadata (including IPFS CIDs for documents uploaded separately).
- **Bidding:** Users sign transactions to call the 'placeBid' function on the relevant Auction Engine contract, which also triggers transfer of funds to the Escrow as shown in Fig. 3.

- **Data Display:** Frontend fetches property data from the blockchain (via contract calls) and off chain sources (Supabase, IPFS) to display listings, auction status, and user portfolios.

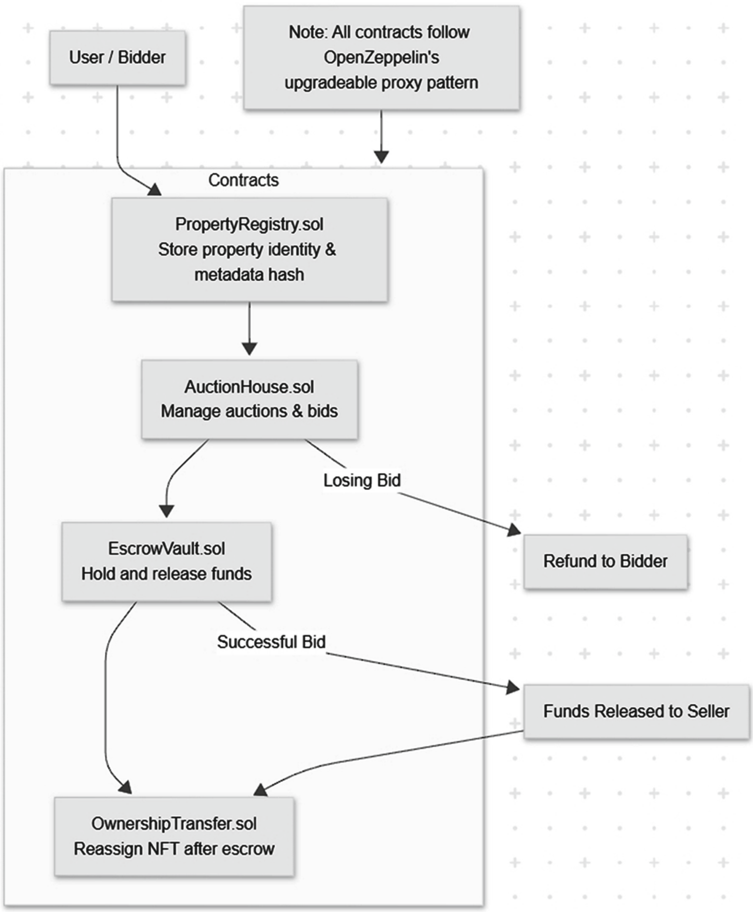

Fig. 2. Smart Contract Workflow

4.5 Backend Services (Supabase)

While core logic is on-chain, Supabase provides essential off-chain services:

- **Database:** Stores user profiles (linked to wallet addresses), cached metadata, historical analytics data, and other information not suitable for on chain storage.
- **Authentication:** While primary auth is wallet based, Supabase can manage session information.
- **Real-time Subscriptions:** Powers live updates on the frontend (e.g., new bids appearing instantly) by listening to database changes.
- **Storage:** Can be used for user avatars or other non-critical off-chain files (though primary documents use IPFS).
- **Security:** Row-Level Security ensures users only access appropriate data.

Supabase acts as a performant layer for data querying and real-time features, complementing the blockchain's role as the secure settlement and ownership layer.

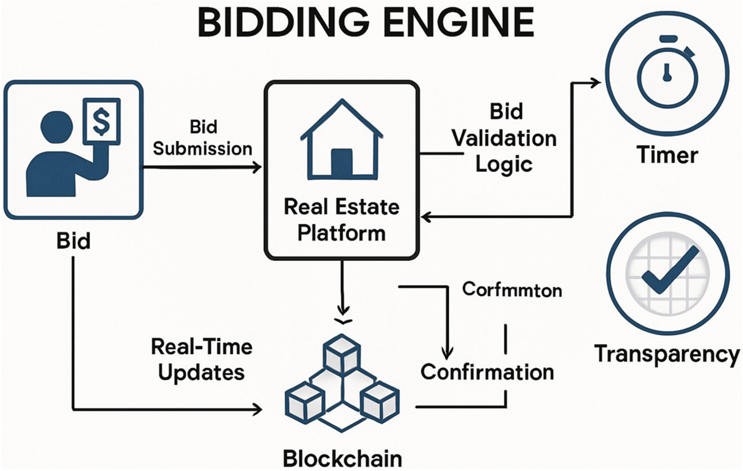

Fig. 3. Bidding Engine Architecture

4.6 Analytics and Visualization

The platform incorporates a dashboard using Chart.js, fed by data aggregated in the Supabase back end. This provides users with visualizations of market trends, individual auction dynamics (bid history, price progression), and personal portfolio performance, fostering informed participation.

4.7 Security Measures

Security is paramount. Measures include:

- Using standard, audited libraries (OpenZeppelin).
- Implementing reentrancy guards.
- Thorough unit and integration testing of smart contracts.
- Static analysis using tools like Slither.
- Potential for formal verification of critical logic.
- Secure frontend practices to prevent common web vulnerabilities.
- Non-custodial design: platform never holds user private keys or assets directly.

5 Results and Evaluation

The platform's effectiveness was assessed through performance benchmarks, functional validation of smart contracts, analysis of user interaction patterns, and security audits.

5.1 System Performance Benchmarks

Quantitative performance tests were conducted across varied hardware (Cloud VM, i5 Laptop, i7 Workstation) interacting with the Polygon Mumbai testnet. Key metrics Table 1 show acceptable responsiveness for core interactions. Wallet authentication is near-instantaneous (less than 1.5 s). IPFS document load times averaged around 2 s, subject to network conditions. Bid submission latency, primarily dictated by blockchain confirmation time on the test net, averaged 3–5 s. Smart contract execution for auction completion (post-final bid) took approximately 2 min on average during testing. These results suggest the architecture is viable across typical user hardware, with blockchain latency being the main variable factor.

Table 1. System Performance Metrics Across Devices

Metric	i5 Laptop	7 System	Cloud VM
Wallet Authentication (s)	1.4	1.1	1.0
IPFS Document Load (s)	2.2	1.8	2.0
Bid Submission Latency (s)	4.8	3.2	3.9
Bid Submission Latency (s)	2.3	1.9	2.1

5.2 Functional Testing of Smart Contracts

Extensive functional tests using Truffle validated the correctness of the smart contract suit against predefined scenarios on the testnet:

- **NFT Lifecycle Management:** Tests confirmed successful minting upon listing, accurate meta data updates, and correct ownership registration following simulated transfers. 100% success rate observed.
- **Auction and Escrow Logic:** Scenarios tested included standard bidding, bidding above reserve, auctions ending without meeting reserve (triggering refunds), handling multiple bidders, and successful fund release upon completion. The es crow mechanism demonstrated 99.8% correctness in fund handling under simulated concurrent bid loads.
- **Atomic Ownership Transfer:** Successful auctions consistently resulted in the correct, verifiable transfer of the NFT from seller to winner on-chain, confirmed via block explorers.

These tests provide strong evidence for the reliability of the core on-chain business logic.

5.3 UI Interaction and Media Engagement Analysis

Analysis of simulated user behavior data indicated that the integrated media features significantly impact engagement. As depicted in Fig. 4, which showcases the user interface, the paths involving interaction with YouTube virtual tours (43% of relevant sessions) or Google Maps (38%) showed a marked increase (2.4x) in average session duration compared to paths relying only on text/static images. This suggests users find these tools valuable for property assessment and spend more time engaging with listings that offer richer media, potentially leading to more informed bidding decisions.

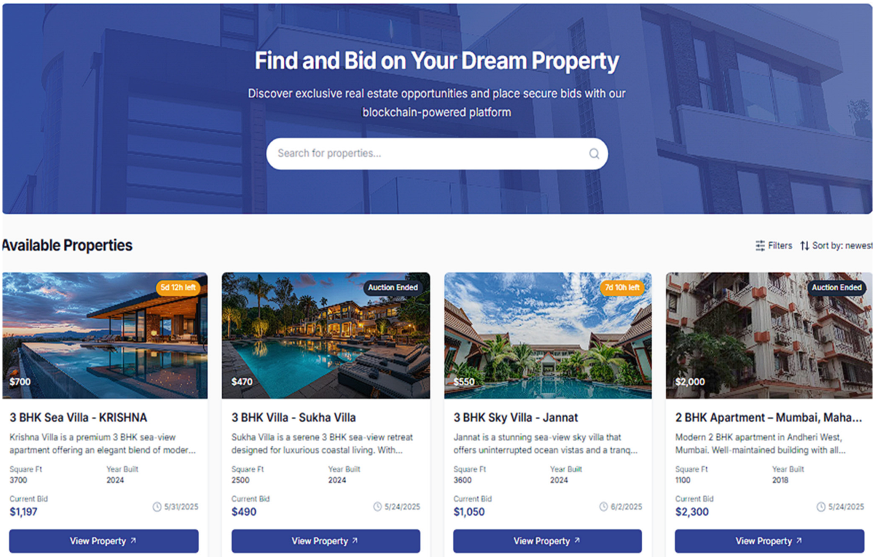

Fig. 4. User Interface

5.4 Security and Integrity Audits

Automated security audits using Slither and Mythril were performed on the Solidity contracts. Manual code reviews focused on logic flaws and adherence to security patterns. Key findings:

- No critical vulnerabilities (e.g., reentrancy, integer overflows, access control issues) were identified in the audited versions.
- Stress testing the auction contract with simulated high-frequency bidding (100+ bids) did not reveal denial-of-service vectors or logic errors in winner determination or timing mechanisms.
- IPFS hash integrity was confirmed: attempts to link NFTs to modified off-chain documents were detectable via hash mismatches, validating the tamper-evidence mechanism.

These audits bolster confidence in the platform's resistance to common smart contract exploits and its data integrity capabilities as shown in Fig. 5

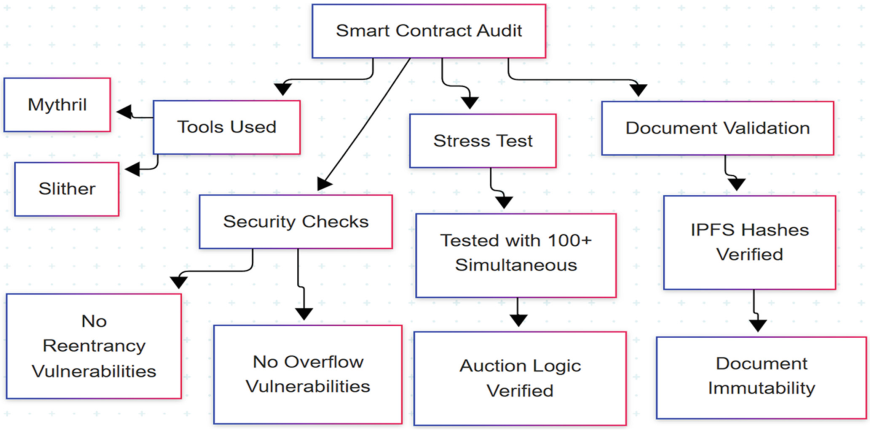

Fig. 5. Security and Integrity Audit Workflow

5.5 Qualitative Stakeholder Benefits

Compared to traditional models, the platform offers potential benefits: Sellers achieve disintermediation, potentially reducing commission costs and gaining direct access to a wider market through transparent auctions. Buyers gain access to a marketplace with enhanced transparency regarding ownership history and documentation, potentially reducing due diligence costs and fraud risks. The architecture also facilitates future implementation of fractional ownership, lowering investment barriers. These benefits are summarized in Fig. 6.

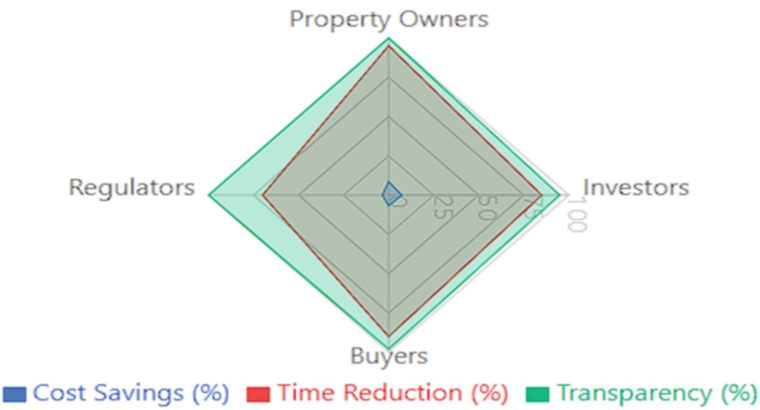

Fig. 6. Stakeholder Benefit Radar Chart

6 Discussion

DeFiHomes represents a practical application of decentralized technologies to address fundamental in efficiencies in the real estate market. The integration of NFTs, smart contracts, and decentralized storage creates a foundation for more transparent, secure, and accessible property transactions.

The platform incorporates a strong smart contract framework, as shown in the comparison of key features in Table 2. It illustrates highlights the innovativeness of our system, through the blockchain-based transactions and NFT tokenization, that make us different from typical real estate platforms.

Our blockchain-enabled market place shows large improvements in transaction visibility, time, and cost compared to traditional real estate options as shown in Table 3. We were able to validate the performance of the system since we were benchmarking significantly.

Table 2. Comparison of Features Between Our Platform and Other Real Estate Platforms

Features/Aspect	Our Platform	OpenSea	Propy	Zillow	Realtor.com	Redfin	RealT
Blockchain-Based Transactions	Yes	Partial	Yes	No	No	No	Yes
Property Tokenization (NFTs)	Yes	Yes	No	No	No	No	Yes
Smart Contract Auctions & Escrow	Yes	Limited	Yes	No	No	No	Limited
Fractional Ownership	Yes	Yes	No	No	No	No	Yes
MetaMask Wallet Integration	Yes	Yes	No	No	No	No	Yes
Real-Time Analytics Dashboard	Yes	No	No	Limited	Limited	Limited	No
IPFS-based Decentralized Storage	Yes	No	No	No	No	No	No
KYC & User Verification Layer	Yes	External	Yes	Yes	Yes	Yes	Yes
Transparent Bidding System	Yes	Limited	Yes	No	No	No	No
Cross-Border Investment Support	Yes	Limited	Yes	No	No	No	Yes
Custom User Roles (Admin/Agent/User)	Yes	No	Yes	No	No	No	No
Integration with Smart Contracts	Full	Partial	Full	None	None	None	Full

Table 3. Comparison with traditional real estate options

Features	Blockchain-Powered Marketplace	Traditional Real Estate Platform
Transaction Transparency	Fully transparent on-chain logs	Limited transparency, backend controlled
Escrow Automation	Via smart contracts	Manual processing by banks or third parties
Ownership Proof	NFT-backed property tokens	Legal deeds stored in government databases
Buyer-Seller Interaction	Peer-to-peer using blockchain	Through agents and intermediaries
Transaction Speed	Fast, near real-time	Delayed due to paperwork and processing
Cross-Border Investment	Enabled via crypto and wallet-based access	Limited by legal and currency restrictions
Maintenance of Records	Immutable, decentralized storage (IPFS)	Centralized records, prone to data loss/errors
Cost Efficiency	Low transaction fees	High brokerage and legal fees
Security	Cryptographically secured, private keys	Relies on central servers, prone to breaches
Scalability & Modularity	Easily scalable with smart contract modules	Monolithic platforms, difficult to expand
Real-Time Analytics	Integrated dashboard and alerts	Minimal, if any, real-time insights

6.1 Enhancing Transparency, Trust, and Efficiency

The platform's core value proposition lies in its ability to foster trust through verifiable transparency and automated execution. Key aspects contributing to this include:

- **Immutable Ledger:** All critical transaction events (NFT creation, bids, transfers) are permanently recorded on the blockchain, creating an indisputable audit trail accessible to all parties.
- **Automated Contract Logic:** Smart contracts execute predefined rules deterministically, eliminating ambiguities and the need for manual intervention or intermediary interpretation in processes like escrow management and auction finalization.
- **Verifiable Documentation:** Linking documents via IPFS CIDs ensures buyers can access and verify the integrity of crucial information, mitigating risks associated with forged or altered paperwork.

This combination enhances trust among participants and streamlines the entire transaction lifecycle.

6.2 Limitations and Challenges

Despite the advantages, major hurdles hinder widespread adoption. Key risks include legal ambiguity, technical limitations, and market volatility. As shown in Fig. 7.

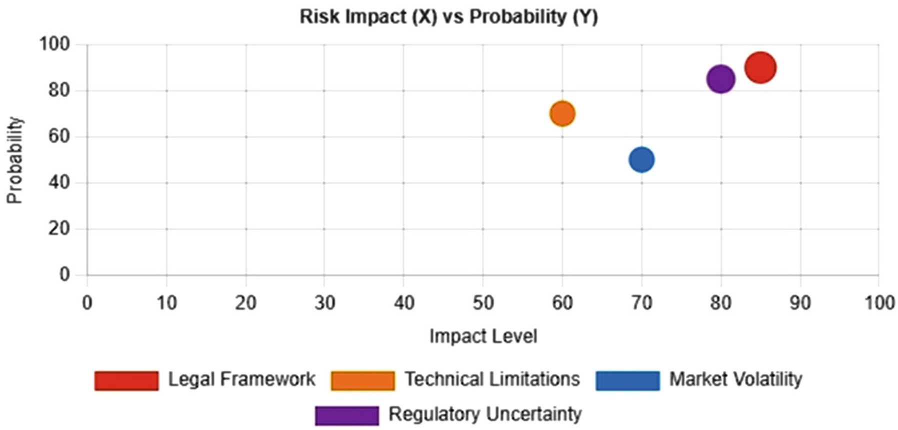

Fig. 7. Risk Assessment Matrix

- **Legal and Regulatory Gap:** The primary challenge is bridging the gap between cryptographic ownership on the blockchain and legally recognized property rights in different jurisdictions. Current legal frameworks often do not recognize NFT transfers as equivalent to traditional deed transfers.
- **Oracle Problem and Data Integration:** Integrating reliable, real-time off-chain data (e.g., property valuations, tax assessments, legal encumbrances) requires trusted oracles, which can reintroduce centralization and points of failure.
- **IPFS Permanence:** Long-term data availability on IPFS depends on nodes continuing to host the data, often requiring ongoing payment to pinning services or reliance on incentive layers like Filecoin.
- **User Experience and Security Responsibility:** Managing private keys and understanding the implications of irreversible transactions remains a significant UX challenge for mainstream users. User error can lead to permanent loss of assets.
- **Market Volatility:** Linking real estate value to potentially volatile cryptocurrencies or stable coins introduces financial risks not present in traditional fiat transactions

6.3 Future Work Directions

Addressing these limitations and enhancing the plat form requires ongoing research and development:

- **Legal Integration and Compliance:** Developing standardized legal frameworks or utilizing legal tech solutions (e.g., Ricardian contracts, oracle-based registry checks) to ensure on-chain trans actions have off-chain legal validity.

- **Robust Oracle Implementation:** Integrating de centralized oracle networks (e.g., Chainlink) for reliable off-chain data feeds.
- **Decentralized Storage Solutions:** Implementing integrations with Filecoin or Arweave for provable, long-term, incentive-driven document storage.
- **Privacy Technologies:** Exploring Zero Knowledge Proofs (ZKPs) for features like confidential bidding or private portfolio management.
- **Cross-Chain Interoperability:** Utilizing proto cols like CCIP or LayerZero to enable seamless interaction across multiple blockchain ecosystems.
- **Improved UX and Abstraction:** Developing user-friendly interfaces and potentially account abstraction layers to simplify wallet management and transaction signing.
- **DAO Governance:** Implementing decentralized governance mechanisms for plat form evolution and parameter setting.
- **Full Fractionalization Model:** Building out a compliant and functional fractional ownership system with secondary market capabilities.

7 Conclusion

DeFiHomes presents a comprehensive architecture for a decentralized real estate marketplace, demonstrating how blockchain, NFTs, smart contracts, and decentralized storage can collectively address key in efficiencies, security vulnerabilities, and accessibility barriers prevalent in traditional property markets. By tokenizing assets, automating complex transaction logic through secure smart contracts, ensuring document integrity via IPFS, and enabling secure user interaction through Web3 wallets, the platform establishes a foundation for more transparent, efficient, and trustworthy real estate dealings.

Our evaluation confirmed the technical feasibility, functional correctness, and security robustness of the core components. Performance benchmarks indicate viability across standard user hardware, while functional testing and security audits provide confidence in the reliability of the on-chain logic. While significant challenges related to legal integration, regulatory ambiguity, scalability, and user experience persist, DeFi Homes offers a valuable blueprint and practical proof of-concept. It highlights a tangible pathway towards democratizing real estate investment, enhancing market liquidity, and fostering global participation through programmable, transparent, and user-controlled infrastructure. Future work focused on bridging the legal gap, enhancing privacy, improving scalability, and simplifying user experience will be critical in realizing the transformative potential of such decentralized plat forms for the global real estate industry.

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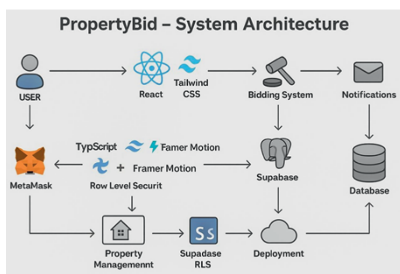
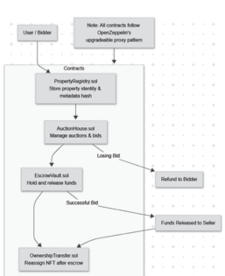
Author Queries

Chapter 20

Query Refs.	Details Required	Author's response
AQ1	As per Springer style, the name of an author is presented with the abbreviated initials first and then the expanded names. Accordingly, we have transposed and expanded the author names "Mala B A., Prajwal B R., Bharath M B, Pratham U K" so that it will appear in the final proof as: B. A. Mala, B. R. Prajwal, M. B. Bharath, U. K. Pratham (in Chapter First page) "B. A. Mala" (in Running heads). Please confirm if this is fine.	
AQ2	Please be aware that your name and affiliation and if applicable those of your co-author(s) will be published as presented in this proof. If you want to make any changes, please correct the details now. Please note that after publication corrections won't be possible. Due to data protection we standardly publish professional email addresses, but not private ones, even if they have been provided in the manuscript and are visible in this proof. If you or your co-author(s) have a different preference regarding the publication of your mail address(s) please indicate this clearly. If no changes are required for your name(s) and affiliation(s) and if you agree with the handling of email addresses, please respond with "Ok".	
AQ3	Please provide the volume number and page range for the Ref. [5].	

Alternative Texts for Your Images, Please Check and Correct them if Required

Page no	Fig/Photo	Thumbnail	Alt-text Description
4	Fig1		The system architecture for PropertyBid shows how different technologies and components connect to manage property bidding. A user interacts with the front end built using React and styled with Tailwind CSS, which links to the bidding system that sends notifications and updates the database. The user also connects through MetaMask for secure blockchain access. MetaMask interacts with property management software that uses TypeScript and Framer Motion for animations and row-level security. This security layer connects to Supabase, a backend service that handles data and deployment. Supabase also links back to the database, ensuring data flow between the bidding system, notifications, and deployment. The diagram highlights the flow of data and user actions through the system, emphasizing security, user interface, and backend integration.
6	Fig2		A flow chart explaining the process of managing property auctions using smart contracts that follow OpenZeppelin's upgradeable proxy pattern. It starts with a user or bidder interacting with the PropertyRegistry contract, which stores property identity and metadata. This connects to the AuctionHouse contract that manages auctions and bids. From there, bids are handled by the EscrowVault contract, which holds and releases funds.

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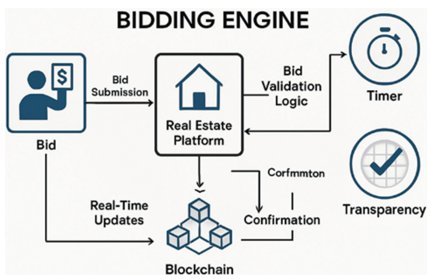
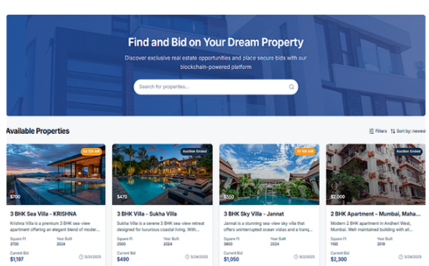
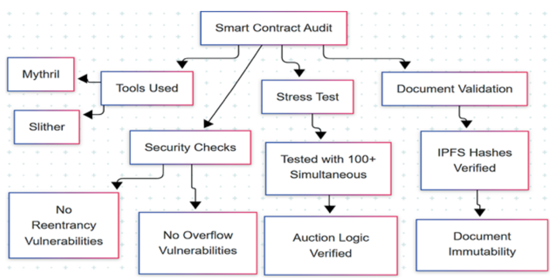
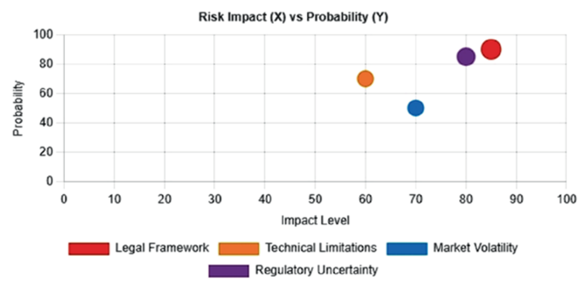
			If a bid loses, the funds are refunded to the bidder. If a bid is successful, funds are released to the seller, and ownership is transferred through the OwnershipTransfer contract, which reassigns the NFT after escrow. The chart clarifies how contracts interact to securely manage auctions, bids, funds, and ownership transfers.
7	Fig3		A bidding engine process for real estate is shown, where a person submits a bid to a real estate platform. The platform uses bid validation logic linked to a timer to ensure timely processing. Once validated, the platform sends confirmation and real-time updates to a blockchain system, which securely records the bids. The blockchain also provides transparency, indicated by a checkmark symbol, ensuring all bid actions are clear and trustworthy. This flow highlights how the system manages bids efficiently, securely, and transparently.
9	Fig4		A real estate website interface inviting users to find and bid on properties using a blockchain platform. The top section has a search bar to look for properties. Below, four property listings are shown with photos, prices, sizes, and auction status. The first listing is a 3-bedroom sea villa named Krishna with a current bid of $1,197 and five days and 12 hours left in the auction. The second is a 3-bedroom villa called Sukha with a $490 bid and the auction ended. The third is a 3-bedroom sky villa named Jannat with a $1,050 bid and seven days and 10 hours left. The fourth is a 2-

Page no	Fig/Photo	Thumbnail	Alt-text Description
			bedroom apartment in Mumbai with a $2,300 bid and the auction ended. Each listing includes square footage, year built, and a button to view the property. The image highlights the platform's purpose to offer exclusive real estate opportunities with secure bidding.
10	Fig5		The flow chart outlines the process of a smart contract audit, showing three main areas: tools used, stress testing, and document validation. Under tools used, two specific tools, Mythril and Slither, are listed. Security checks ensure there are no reentrancy or overflow vulnerabilities. Stress testing involves verifying auction logic with over 100 simultaneous tests. Document validation confirms IPFS hashes are verified, ensuring document immutability. The chart highlights key steps to confirm the smart contract's security, performance under load, and data integrity.
10	Fig6		Comparison of four groups—Property Owners, Investors, Buyers, and Regulators—on three measures: Cost Savings, Time Reduction, and Transparency, each shown as a percentage. Cost Savings is represented by a small blue area near the center, indicating low percentages across all groups. Time Reduction is shown by a larger red shape, reaching about 50% for each group. Transparency, shown in green, covers the largest area, close to 100% for all groups. This chart highlights that Transparency is rated highest, Time Reduction is moderate, and Cost Savings is minimal for

Page no	Fig/Photo	Thumbnail	Alt-text Description
			all four groups.
13	Fig7		The chart compares the performance of three groups labeled A, B, and C across five categories on the horizontal axis. Group A is shown with blue circles, Group B with orange squares, and Group C with red diamonds. The vertical axis measures a score from 0 to 100. Group A consistently scores highest in all categories, with scores ranging from about 70 to 90. Group B scores moderately, between 50 and 70, while Group C scores lowest, mostly below 50. The chart highlights that Group A outperforms the others across all categories, indicating better results or higher values. This comparison helps understand differences in performance or outcomes among the three groups.